

Derivatives — group rounds

Lessons 1-4, 7-12 · Quarter I · 40 min · 20 questions · 100 points · groups of 3-4

Four rounds, each timed and worth more than the last. The tag on every question names the topic it comes from. Work on paper, hand in one sheet for the group. An answer with no method scores half.

LESSON CLOCK · 40 MINUTES

4'	5'	8'	9'	8'	6'
Form groups, read the rules	Round A	Round B	Round C	Round D	Answers on the board, scores

1. Groups of 3-4. One answer sheet per group, with every member's name on it.
2. Rounds are timed and called by the teacher. When a round is called, pens down and move on — an unfinished round scores what it has.
3. In rounds C and D any member may be asked to explain the method. If nobody can, the group scores half for that question, however correct the answer is.
4. No calculators. Leave surds and π exact, and factorise every derivative.

Round A · Easy

5'

2 pts each

Two from each topic — speed round

1

T1

For $y = x^2$, write the increment Δy as x goes from 3 to $3 + h$.

$6h + h^2$
 $(3 + h)^2 - 9 = 9 + 6h + h^2 - 9.$

2

T1

Find the slope of the secant to $y = \frac{1}{x}$ between $x = 1$ and $x = 2$.

$-\frac{1}{2}$
 $\frac{\frac{1}{2} - 1}{2 - 1} = -\frac{1}{2}.$

3

T2

Evaluate $\lim$ as $x \rightarrow 3$ of $\frac{x^2 - 9}{x^2 - 2x - 3}$.

$\frac{3}{2}$
Both factorise through $(x - 3)$: $\frac{(x - 3)(x + 3)}{(x - 3)(x + 1)} = \frac{x + 3}{x + 1} \rightarrow \frac{6}{4}.$

4

T2

Evaluate $\lim$ as $x \rightarrow 0$ of $\frac{\sqrt{x + 1} - 1}{x}$.

$\frac{1}{2}$
Conjugate: the numerator becomes x , leaving $\frac{1}{\sqrt{x + 1} + 1}.$

5

T3

Differentiate $y = \frac{2x^3 - 5}{x^2}$.

$2 + \frac{10}{x^3}$
Divide first: $y = 2x - 5x^{-2}$, so $y' = 2 + 10x^{-3}.$

6

T3

Differentiate $y = x\sqrt{x}$.

$\frac{3}{2}\sqrt{x}$
 $x\sqrt{x} = x^{3/2}$, so $y' = \frac{3}{2}x^{1/2}.$

7

T4

Differentiate $y = \frac{1}{(2x - 7)^3}$.

$-\frac{6}{(2x - 7)^4}$
 $y = (2x - 7)^{-3}$, so $y' = -3(2x - 7)^{-4} \times 2.$

8

T4

Differentiate $y = \sqrt[3]{3x + 1}$.

$\frac{1}{(3x + 1)^{2/3}}$
 $y = (3x + 1)^{1/3}$, so $y' = \frac{1}{3}(3x + 1)^{-2/3} \times 3$ — the 3s cancel.

Round B · Medium

8'

4 pts each

Rewrite before you differentiate

9

T1

Write the secant slope of $y = \sqrt{x}$ between $x = 4$ and $x = 4 + h$ without a surd in the numerator, then state its limit as $h \rightarrow 0$.

$\frac{1}{\sqrt{(4 + h) + 2}} \rightarrow \frac{1}{4}$
Multiply above and below by $\sqrt{(4 + h) + 2}$: the numerator becomes $(4 + h) - 4 = h$, which cancels the h below.

10

T2

Evaluate $\lim$ as $x \rightarrow 2$ of $\frac{x^3 - 8}{x - 2}$.

12
 $x^3 - 8 = (x - 2)(x^2 + 2x + 4)$, so the limit is $4 + 4 + 4.$

11

T2

Evaluate $\lim$ as $x \rightarrow 1$ of $\frac{1}{x - 1} - \frac{2}{x^2 - 1}$.

$\frac{1}{2}$
Neither part has a limit on its own. Over a common denominator:
 $\frac{(x + 1) - 2}{(x - 1)(x + 1)} = \frac{x - 1}{(x - 1)(x + 1)} = \frac{1}{x + 1}.$

12

T3

Differentiate $y = (x + 1)(x - 2)(x + 3)$.

$3x^2 + 4x - 5$
Expanding is quicker than two product rules: $y = x^3 + 2x^2 - 5x - 6.$

13

T3

Differentiate $y = \frac{x^2 + 1}{x^2 - 1}$.

$-\frac{4x}{(x^2 - 1)^2}$
Quotient rule; the numerator is $2x(x^2 - 1) - (x^2 + 1)2x = 2x(-2) = -4x.$

14

T4

Differentiate $y = (1 - 2x)^6(x + 3)$ and factorise.

$-7(1 - 2x)^5(2x + 5)$
Product rule with the chain rule on the first factor:
 $-12(1 - 2x)^5(x + 3) + (1 - 2x)^6$. Take out $(1 - 2x)^5$; what is left is $-12(x + 3) + (1 - 2x) = -14x - 35.$

15 T1 T2 For $f(x) = \frac{1}{x^2}$, write the secant slope between $x = 2$ and $x = 2 + h$ as a single fraction, simplify it, and find its limit as $h \rightarrow 0$.

$$-\frac{4+h}{4(2+h)^2} \rightarrow -\frac{1}{4}$$

Over a common denominator the numerator is $4 - (2+h)^2 = -4h - h^2 = -h(4+h)$, and the whole quotient is $\frac{-h(4+h)}{4(2+h)^2 h}$.

The h cancels; then $h \rightarrow 0$ gives $-\frac{4}{16}$.

16 T2 Evaluate $\lim_{x \rightarrow 4} \frac{\sqrt{x}-2}{x^2-16}$.

$$\frac{1}{32}$$

Factorise the denominator as $(x-4)(x+4)$ and multiply above and below by $\sqrt{x}+2$. The numerator becomes $x-4$, which cancels, leaving

$$\frac{1}{(\sqrt{x}+2)(x+4)} \rightarrow \frac{1}{4 \times 8}$$

17 T3 The curve $y = \frac{x^2-1}{x^2+1}$ has a horizontal tangent. Find where, and the value of y there.

$x = 0$, where $y = -1$

Quotient rule: the numerator is $2x(x^2+1) - (x^2-1)2x = 4x$, so $y' = \frac{4x}{(x^2+1)^2}$. A fraction is zero only when its numerator is, and the denominator is never zero — so $x = 0$ alone.

18 T4 Differentiate $y = \sqrt{\frac{2x+1}{2x-1}}$ for $x > \frac{1}{2}$.

$$-\frac{2}{(2x-1)\sqrt{4x^2-1}}$$

Chain rule on $u^{1/2}$ with $u = \frac{2x+1}{2x-1}$. The quotient rule gives $\frac{du}{dx} = \frac{2(2x-1) - 2(2x+1)}{(2x-1)^2} = -\frac{4}{(2x-1)^2}$, so $y' = \frac{1}{2\sqrt{u}} \times (-\frac{4}{(2x-1)^2})$. Writing $\sqrt{u} = \frac{\sqrt{2x+1}}{\sqrt{2x-1}}$ and tidying gives the answer. The derivative is negative everywhere on the domain, which matches a decreasing curve.

Round D · Very hard 8' 16 pts each

Every part needed for full marks

19 T3 T4 (a) Show that $y = \frac{x}{\sqrt{(x^2+4)}}$ has $y' = \frac{4}{(x^2+4)^{3/2}}$. (b) Hence find the equation of the tangent at $x = 0$. (c) Explain why the curve has no stationary point.

(b) $y = \frac{x}{2}$ (c) y' is never zero

(a) Quotient rule: $[\sqrt{(x^2+4)} - x \cdot \frac{x}{\sqrt{(x^2+4)}}] / [(x^2+4)]$. Multiply above and below by $\sqrt{(x^2+4)}$: the numerator becomes $(x^2+4) - x^2 = 4$ and the denominator $(x^2+4)^{3/2}$.

(b) $y(0) = 0$ and $y'(0) = \frac{4}{8} = \frac{1}{2}$, so the tangent is $y = \frac{1}{2}x$.

(c) The numerator of y' is the constant 4, so $y' \neq 0$ for every x — the curve is increasing everywhere and never levels off.

20 T1 T3 A stone is thrown upwards and its height is $h(t) = 25t - 5t^2$ metres after t seconds.

(a) Write the average velocity between $t = 1$ and $t = 1 + k$ as an expression in k , simplified. (b) State its limit as $k \rightarrow 0$, and check it against $h'(1)$ found by the rules. (c) Find the greatest height the stone reaches.

(a) $15 - 5k$ (b) 15 m/s , and $h'(1) = 15 \checkmark$ (c) $31,25 \text{ m}$

(a) $h(1) = 20$ and $h(1+k) = 20 + 15k - 5k^2$, so the average is $\frac{15k - 5k^2}{k} = 15 - 5k$.

(b) As $k \rightarrow 0$ this tends to 15. By the rules $h'(t) = 25 - 10t$, so $h'(1) = 15$ — the average velocity over a shrinking interval and the rate at the instant agree, which is the whole point of lessons 1–2.

(c) The stone is at its highest when $h'(t) = 0$, that is $t = 2,5 \text{ s}$, and $h(2,5) = 62,5 - 31,25 = 31,25 \text{ m}$.

TOPIC COVERAGE

TOPIC	QUESTIONS	WHICH
T1 Increments and the problem of the tangent · lessons 1–2	5 1, 2, 9, 15, 20	
T2 The limit of a function · lessons 3–4	6 3, 4, 10, 11, 15, 16	
T3 The rules of differentiation · lessons 7–9	7 5, 6, 12, 13, 17, 19, 20	
T4 The derivative of a composite function · lessons 10–12	5 7, 8, 14, 18, 19	

SCORE

ROUND	QUESTIONS	EACH	ROUND TOTAL
Round A · Easy	8	2	16
Round B · Medium	6	4	24
Round C · Hard	4	7	28
Round D · Very hard	2	16	32
Total	20		100